



OPEN Teaching activity design and psychological practice of music majors under a convolutional neural network and transformer module

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The purpose of this study is to explore a new mode of teaching activity design and psychological practice for music majors supported by the concept of STEAM education. In this study, a personalized recommendation system based on a convolutional neural network (CNN) and a Transformer module is designed and implemented. The STEAM education concept is further introduced, which is combined with a personalized recommendation system to design music teaching activities to improve students' learning effect and comprehensive quality. Finally, the performance of the model is evaluated experimentally. The results show that the accuracy, recall, and F1 values of the personalized recommendation system designed here reach 95.76%, 90.28%, and 92.41%, respectively, which is more than 4% higher than other reference models. And in the user behavior indicators, the number of clicks, browsing time, and comments has increased by more than 10%, which reveals the remarkable advantages of the proposed algorithm in improving teaching effect and learning experience. Therefore, the proposed CNN-Transformer algorithm shows obvious advantages in prediction accuracy and user participation, which provides new empirical support for the application of educational technology in music education.

Keywords STEAM education concept, Music major, Deep learning, Transformer, Personalized recommendation system

Research background and motivations

As an interdisciplinary educational concept, STEAM education emphasizes the comprehensive application of Science, Technology, Engineering, Art, and Mathematics to cultivate students' innovative thinking and problem-solving ability^{1–4}. In the field of music education, the introduction of the STEAM education concept not only enriches the teaching content but also promotes the development of students' various abilities^{5,6}. In recent years, with the rapid development of information technology, machine learning has been widely used in the field of education, especially in personalized recommendation learning systems. A personalized recommendation learning system can provide customized learning resources and teaching activities according to students' personal characteristics, interests, and learning needs, thus improving learning efficiency and learning effect^{7–9}. However, how to effectively apply machine learning technology in music education^{10,11}, especially how to design and implement teaching activities suitable for music majors through a personalized recommendation learning system, is an urgent problem to be studied.

Although machine learning has been applied in music education, existing research predominantly relies on traditional convolutional neural networks (CNNs) or simple collaborative filtering algorithms. These methods typically capture only local spectral features of musical audio (such as timbre or momentary rhythms), while overlooking the inherent long-term dependencies and global context information of music as time-series data. Within the STEAM education context, precise instructional recommendations demand a deeper semantic understanding of musical content. Therefore, overcoming the limitations of traditional CNNs in modeling long sequences and integrating advanced attention mechanisms to enhance the accuracy and interpretability of recommendation systems represents critical technical challenges that require urgent resolution.

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Research objectives and contributions

The purpose of this study is to explore the application of teaching activity design and psychological practice of music majors based on STEAM education with the support of artificial intelligence (AI) technologies such as deep learning. By combining machine learning technology with the idea of STEAM education, a personalized recommendation learning system that can dynamically adapt to students' needs is constructed to provide a more efficient and effective learning experience for music majors.

The core innovation of this study lies not in proposing an entirely new foundational network architecture, but in designing a cross-modal feature fusion mechanism tailored to the specific context of music education. This study constructs a three-dimensional integrated framework encompassing “technology-pedagogy-psychology,” systematically exploring innovative music education models under the STEAM philosophy. Unlike generic recommendation models, music educational materials simultaneously embody acoustic attributes (timbre, rhythm, dynamics) and pedagogical attributes (musical structure, harmonic progression, technical difficulty), each requiring fundamentally distinct feature representations. The innovative contributions of this study manifest in three aspects: (1) Dual-path feature extraction: CNNs capture local spatiotemporal textures for the acoustic modality, while Transformers model global structural dependencies for the symbolic modality—a division aligned with music cognition's dual nature; (2) Feature fusion strategy: Independent optimization of both modalities before the fusion layer, rather than simple concatenation, ensures their feature spaces remain mutually independent; (3) Instruction-oriented symbolic feature engineering: Quantifying pedagogical attributes like technical difficulty metrics and harmonic complexity into learnable feature vectors, endowing the recommendation system with “pedagogical awareness.”

Literature review

STEAM education is an interdisciplinary education concept that has emerged in recent years. It aims to cultivate students' innovative thinking and comprehensive problem-solving ability by integrating science, technology, engineering, art, and mathematics. Many scholars have done relevant research. Belbase et al. (2022)¹² discussed the prospects, priorities, processes, and problems of STEAM education and emphasized the importance of multidisciplinary integration in cultivating innovative and critical thinking. Dignam (2023)¹³ analyzed the teaching of professional knowledge in STEAM education and discussed how interdisciplinary methods could improve students' academic ability and practical skills. Gan & Bai (2023)¹⁴ combined Marx's thoughts of critical pedagogy, reproduction, and aesthetic literacy to explore its application in STEAM education and its enlightenment to educational practice. Mercan & Kandır (2024)¹⁵ studied the influence of the early STEAM education program on children's visual-spatial reasoning ability, and the results showed that the program significantly improved children's related skills. Perales & Aróstegui (2024)¹⁶ discussed the implementation of STEAM education and its educational, social, and economic impact, and emphasized the advantages of this method in the cultivation of comprehensive quality. Huang & Qiao (2024)¹⁷ improved the computational thinking skills of STEAM high school students through artificial intelligence (AI) education, which proved the effectiveness of AI education in cultivating students' ability to solve complex problems. Jeong & González-Gómez (2024)¹⁸ discussed the use of positive and innovative methods to promote the sustainable development of STEAM and mathematics education, and emphasized the importance of students' active learning and innovative ability.

Music education is an important part of art education, and how to improve the teaching effect by using machine learning technology is a hot research topic in recent years. Yu et al. (2023)¹⁹ summarized the development and application of AI in music education, discussed the potential of AI technology in personalized learning and music creation, and improved students' learning experience and teaching effect. Li & Sun (2023)²⁰ discussed the innovative application of intelligent technology in music and aesthetic education courses. Through machine learning and big data analysis, the interactivity of curriculum design and students' aesthetic ability was improved. Wang (2023)²¹ studied the application of machine learning technology in classical music education, focusing on the effectiveness of machine learning in music style classification and track recommendation, which improved the teaching effect. Xu (2024)²² discussed the development of an intelligent electronic learning system based on digital audio technology in preschool music education, which showed that the system had a remarkable effect in improving children's music literacy. Cui & Chen (2024)²³ proposed a new vocal music education framework combining a convolutional neural network (CNN) and diversified learning methods, which proved its effectiveness in improving vocal music teaching quality and students' learning effect. Acevedo & Mendoza (2024)²⁴ introduced a music software based on a machine learning feedback system, as an alternative to children's elementary piano learning, showing its potential in improving children's learning interest and skills. However, most of the research has been limited to using a standard CNN for audio feature extraction. While CNNs excel at capturing local spatial features (such as textures in spectrograms), their restricted receptive fields make it challenging to effectively handle long-range temporal correlations in musical signals. In complex music education scenarios, the lack of global context modeling often leads to biased recommendation outcomes. In recent years, the Transformer architecture has demonstrated superior performance to traditional CNNs in sequence modeling tasks, yet its application in music education recommendation systems remains in its infancy.

Personalized recommendation learning system has been widely studied in the field of education in recent years. A personalized recommendation system can provide customized learning resources and activities for students by analyzing students' learning behavior, learning style, and hobbies, and improve learning effect and interest. Li et al. (2023)²⁵ proposed a personalized prompt learning method, which was used in the interpretive recommendation system. By enhancing the interpretability of the model, the accuracy and user satisfaction of personalized recommendations were improved. Lim et al. (2023)²⁶ discussed the influence of personalized scaffolding based on real-time analysis on students' self-regulated learning, and the results showed that personalized scaffolding significantly improved students' learning autonomy and effect. Safarov et al. (2023)²⁷, combined with deep learning technology, optimized the recommendation system in e-education through

clustering and sequence analysis, which significantly improved learners' personalized learning experience and learning effect. Huang et al. (2023)²⁸ evaluated the influence of AI-driven personalized recommendations on learners' participation, motivation, and learning achievement in the flip classroom and found that the AI recommendation system could effectively improve learners' learning effect. Raj & Renumol (2024)²⁹ proposed an improved adaptive learning path recommendation model based on real-time learning analysis, which proved that it had significant advantages in the recommendation and optimization of personalized learning paths. Chen et al. (2024)³⁰ applied a machine learning algorithm to optimize a personalized education recommendation system, and improved the practicability and user satisfaction of the education recommendation system by enhancing the accuracy and efficiency of the recommendation algorithm.

In summary, although existing literature has explored the integration of STEAM education concepts and personalized recommendation systems in music education, significant limitations remain. First, at the algorithmic level, most existing studies employ basic CNN models, which struggle to capture the complex temporal characteristics of music data. This results in insufficient recommendation accuracy and personalization, lacking integration with state-of-the-art (SoTA) deep learning technologies such as Transformers. Second, empirical studies lack rigorous benchmark-based experimental validation, limiting the persuasiveness of research findings.

Therefore, this study pioneers the integration of Transformer modules into music teaching recommendation systems through algorithmic design, constructing a CNN-Transformer hybrid model that addresses traditional models' neglect of global dependencies. Rigorous comparative testing against multiple mainstream algorithms provides substantial numerical evidence. Furthermore, by deeply integrating this high-performance model with STEAM teaching activities and validating its effectiveness through practical teaching feedback, the study demonstrates its ability to enhance students' psychological well-being and comprehensive capabilities. This fills a gap in the field of technology application and pioneers a new paradigm for empirical research in educational technology.

Research model

Demand analysis of the STEAM education concept in music teaching

STEAM education is an interdisciplinary education method that integrates science, technology, engineering, art, and mathematics, and cultivates students' innovative thinking, problem-solving ability, and comprehensive quality through project-based learning activities^{31–33}. It emphasizes practical problem-oriented learning, encourages students to apply knowledge in an interdisciplinary way when solving practical challenges, and cultivates their critical thinking and practical ability³⁴. At present, there are some disadvantages in music teaching, such as a single teaching method, unsatisfied individual needs, disjointed theory and practice, and insufficient psychological support. It lacks sufficient interactivity and students' active participation, which cannot effectively stimulate learning interest and solve psychological problems. Therefore, it is of great significance to introduce the idea of STEAM education into music teaching. Through interdisciplinary integration, project-based learning, and a personalized support system, the teaching quality can be improved, and students' all-round development and innovation ability can be cultivated. The demand is shown in Fig. 1.

In Fig. 1, in view of the current situation of music teaching, after introducing the concept of STEAM education, music is combined with knowledge of science, technology, engineering, and mathematics to help students form a comprehensive knowledge system and improve their comprehensive quality. Project-based and problem-oriented learning methods can stimulate students' interest in learning, increase classroom participation, and promote deep learning. With the application of a personalized recommendation system, teaching can be adjusted according to students' interests and abilities, and customized learning resources and activities can be provided to meet the learning needs of different students. In addition, STEAM education emphasizes innovation and critical thinking, which can effectively cultivate students' creativity and independent thinking ability. Integrating psychological support into teaching activities can help students overcome psychological problems such as stage fright and practice anxiety, and improve their psychological quality and self-confidence.

Through a detailed understanding of the needs of music majors in STEAM education, this study aims to build a personalized recommendation learning system based on deep learning to realize the personalization and intelligence of music teaching, meet the diverse needs of students, and improve their learning effect. A personalized recommendation learning system provides customized learning resources and activities according to students' learning behaviours, interests, and abilities, maximizes the advantages of STEAM education, and promotes students' all-round development.

Construction and analysis of a personalized recommendation learning system model based on deep learning

To address the personalized learning needs of music majors, this study constructs a multimodal deep learning framework integrating acoustic information with symbolic musical information. Unlike unimodal approaches relying solely on audio signals, this model recognizes that professional music education requires understanding music both as sound and as structured teaching material. Therefore, this study employs CNNs^{35–37} for audio feature extraction and introduces an MHSA-based Transformer module^{38–40} to process symbolic musical representations. This approach establishes a CNN-Transformer-based personalized music learning recommendation system model, as illustrated in Fig. 2.

In Fig. 2, the model first collects and preprocesses two distinct types of user behavior data and music content data from the learning platform:

(1) Acoustic modes. Include audio signals of musical works involving user interactions (browsing, listening, and commenting). Audio data is converted into Mel-spectrograms to capture timbre and temporal characteristics. Pre-processing operations such as cleaning, de-duplication, and normalization are carried out on the collected data to ensure the data quality. Data normalization is shown in Eq. (1):

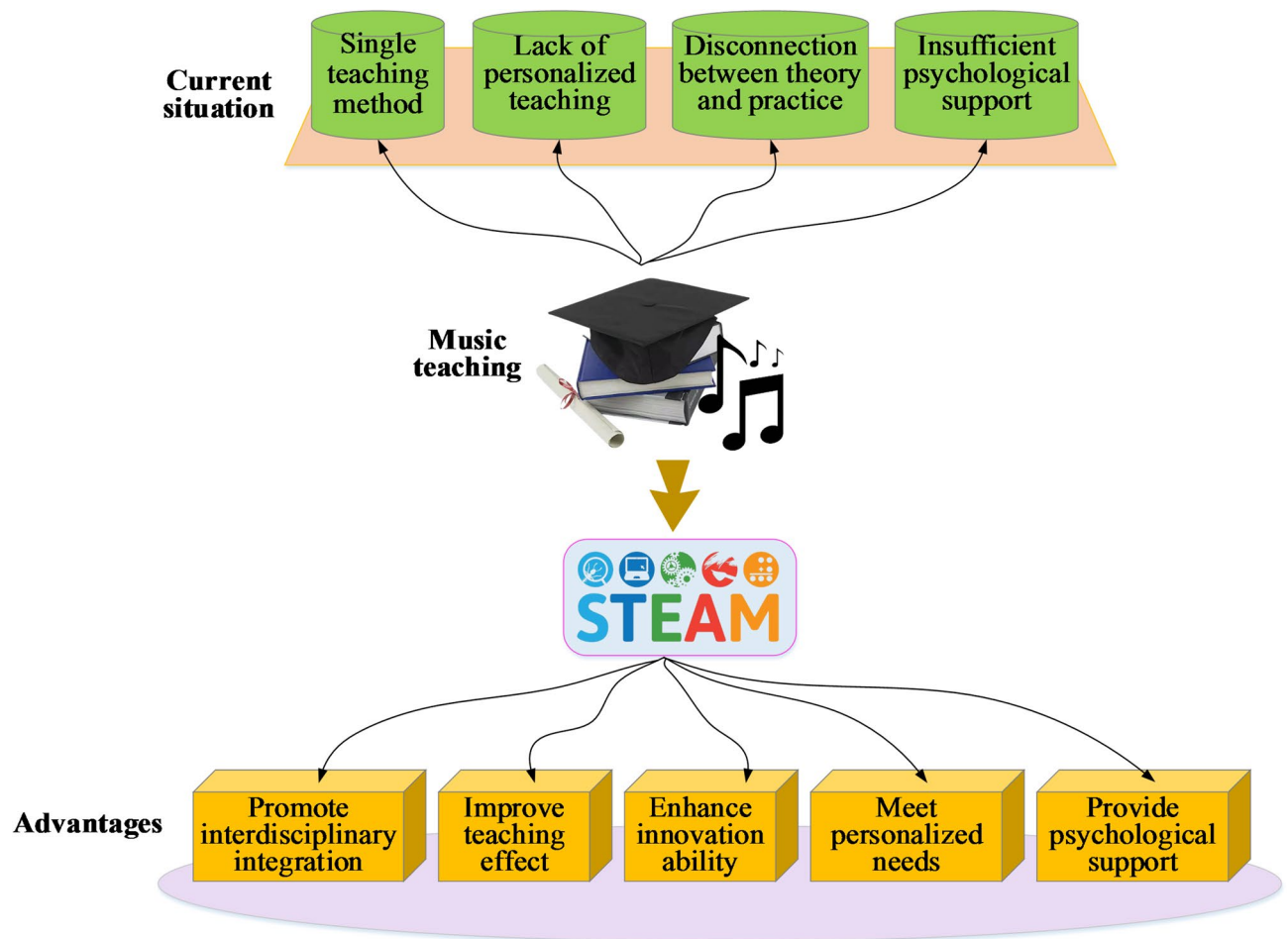


Fig. 1. The present situation of music teaching and the necessary schematic diagram of the STEAM education concept support.

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (1)$$

x refers to the original data, x' refers to the normalized data. $\min(x)$ and $\max(x)$ represent the minimum value and maximum value of the data.

(2) Symbolic modality. This modality captures the essence of music pedagogy essential to the professional development of music students, including: - Score Information: Sheet music encoded in musical instrument digital interface (MIDI) format, containing notes, rhythms, dynamics, and performance markings. - Harmonic Structure: Chord progressions, tonality, and harmonic complexity extracted from symbolic representations. Technical Difficulty Metrics: Quantitative indicators of technical demands, including note density, tempo changes, hand independence requirements, and presence of advanced techniques (e.g., rapid arpeggios, complex fingerings). Music Theory Annotations: Labels for the form, stylistic period, and theoretical concepts embodied in the piece. Symbolic data undergoes preprocessing through tokenization and embedding, converting discrete musical elements into continuous vector representations suitable for deep learning models.

(3) Psychological modality. To integrate psychological practices into the recommendation system, this study incorporates a psychological feature collection module. Standardized psychological scales are used to gather students' psychological state data, including: - Kenny Music Performance Anxiety Scale: Assesses anxiety levels before stage performances and classroom presentations. - General Self-Efficacy Scale: Measures confidence levels in music learning. The Self-Reported Anxiety Scale for Learning assesses anxiety levels during daily practice and coursework. After standardization, psychological data are integrated with acoustic and symbolic features to form multimodal inputs, enabling the recommendation system to perceive students' psychological states and perform adaptive adjustments.

(4) STEAM dimension tags. To enable precise recommendations for interdisciplinary resources, this study annotated each teaching piece with its associated STEAM dimension tags. The annotation system comprises five dimensions: Science: Works involving physical acoustic knowledge such as acoustic principles, vibration frequency, and harmonic series; Technology: Works involving technological applications like electronic music, digital audio processing, and music programming; Engineering: Works involving engineering concepts

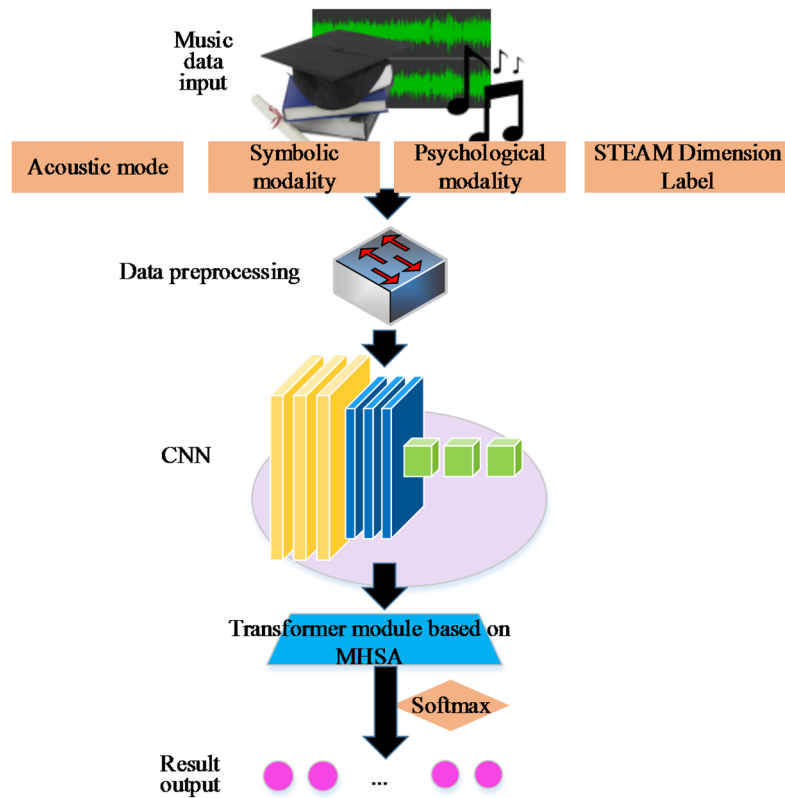


Fig. 2. Schematic diagram of a multi-modal personalized music learning recommendation system based on CNN-Transformer.

like instrument construction, sound field design, and recording engineering; Arts: Core artistic dimensions including musical expressiveness, aesthetic interpretation, and emotional expression; Mathematics: Works involving mathematical principles such as rhythmic proportions, harmonic mathematical relationships, and formal symmetry. Each piece may be associated with multiple dimensions simultaneously—for example, “acoustic principles in electronic music” would be tagged under both Science and Technology—forming a multidimensional tag vector.

Secondly, the CNN algorithm is used to extract data features. Let $X_t = \{x_1, x_2, \dots, x_T\}$, $X_t \in R^{F \times M \times T}$ refers to the training dataset. For audio data, F refers to the frame number of Mel spectrogram of audio data. M refers to the number of filters, and T refers to the number of training samples. Then the supervised timbre feature g of the t -th training sample connects the outputs of each convolution block by using a connection operation, which is expressed as Eq. (2) to (4):

$$Fra_t = \text{concatenate}(y_1, y_2, \dots, y_j) \quad (2)$$

$$y_j = D_j(x_t) \quad (3)$$

$$D_j(x) = \text{pooling}(W_j x + b_j) \quad (4)$$

x_t refers to the t -th training sample, $t = 1, 2, \dots, T$. *pooling* refers to the pooling operation. *concatenate* refers to connection operation. W_j refers to the convolution kernel of the convolution layer in the j -th convolution block. y_j refers to the j -th output of the tone module. b_j refers to the j -th deviation of the tone module. D_j refers to the j -th convolution.

In the supervised time module, after the convolution process of all convolution layers is completed, the outputs of all convolution layers are connected to form a supervised timbre feature, as shown in Eq. (5) to (7):

$$Fre_t = \text{concatenate}(\tilde{y}_1, \tilde{y}_2, \dots, \tilde{y}_k) \quad (5)$$

$$\tilde{y}_k = W_k \tilde{x}_t + b_t \quad (6)$$

$$\tilde{x}_t = \text{pooling}(x_t) \quad (7)$$

In Eq. (5) to (7), Fre_t denotes the supervised temporal feature of the t -th training sample, and *concatenate* is the concatenation of the outputs from all convolutional layers. $\tilde{y}_k$ denotes the output of the k -th layer of the supervision time module. $\tilde{x}_t$ refers to the output result of the pooling operation on the training sample x_t .

Finally, to capture long-range global dependencies within musical feature sequences, this study incorporates a Transformer module. First, the feature map $\tilde{x}_t$ extracted by the CNN is reshaped into a feature sequence $X \in R^{L \times d}$, where L denotes the sequence length and d represents the feature dimension. Subsequently, the MHSA calculates correlations between features. For the input sequence X , the query matrix (Q), key matrix (K), and value matrix (V) are computed as shown in Eq. (8):

$$Q = XW^Q, K = XW^K, V = XW^V \quad (8)$$

W^Q, W^K, W^V denotes the learnable weight matrix. Finally, the attention weights are computed using a scaled dot-product and weighted summation to obtain a feature representation incorporating global contextual information, as shown in Eq. (9):

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d^k}})V \quad (9)$$

$\sqrt{d^k}$ is the scaling factor used to prevent gradient vanishing. Softmax refers to the softmax function (softmax function), where QK^T denotes the product of the query matrix and the transposed key matrix.

Through the multi-head mechanism, the model can learn information from different representational subspaces:

$$g_t^{symbol} = Concat(head_1, \dots, head_h)W^o \quad (10)$$

g_t^{symbol} denotes the symbol feature of the t -th training sample, $head_i$ denotes the i -th attention head, h denotes the total number of attention heads, and W^o is the output projection matrix.

After extracting acoustic and semantic features, where g_t^{audio} denotes the acoustic features of the t -th training sample, the psychological feature vector g_t^{psych} serves as a moderator for recommendation outcomes. This study employs a feature-level fusion strategy, concatenating feature vectors from three modalities:

$$g_t^{fusion} = Concat(g_t^{audio}, g_t^{symbol}, g_t^{psych}) \quad (11)$$

g_t^{fusion} denotes the fusion feature of the t -th training sample. g_t^{psych} refers to the psychological feature vector of the t -th training sample, encompassing psychological indicators such as anxiety level and self-efficacy.

The merged feature vector g_t^{fusion} incorporates both the acoustic characteristics of the music and its pedagogical and psychological attributes, reflecting both the piece's auditory aesthetics and its structured information as teaching material. Through this approach, the model dynamically adjusts its recommendation strategy based on the student's psychological state. For instance, it prioritizes recommending teaching pieces with moderate technical difficulty and strong emotional support for highly anxious students, while suggesting beginner-level works that facilitate success experiences for those with low self-efficacy. This approach meets learning needs while providing psychological support. Subsequently, the merged features undergo dimensionality reduction and nonlinear transformation through a fully connected layer, ultimately outputting recommendation probabilities via a softmax function:

$$\hat{y}_t = softmax(FC(g_t^{fusion})) \quad (12)$$

$\hat{y}_t$ denotes the predicted label of the t -th training sample, and FC refers to the fully connected layer. The cross-entropy function is employed as the model's loss function, as shown in Eq. (13):

$$L = -\frac{1}{T} \sum_{t=1}^T \sum_{c=1}^C y_{t,c} \log(\hat{y}_{t,c}) \quad (13)$$

L denotes the loss function value, C denotes the number of classes, $y_{t,c}$ represents the true label of the t -th training sample in the c -th class, and $\hat{y}_{t,c}$ represents the predicted probability of the t -th training sample in the c -th class.

Through this approach, the model not only learns behavioral weights reflecting users' preferences for musical timbre but also gains a deeper understanding of the pedagogical attributes and technical difficulty of musical works. This enables it to effectively identify and recommend personalized music learning resources that align with students' auditory preferences while supporting their professional development.

To deeply integrate STEAM education principles into the recommendation system, this study designed a dimension-aware interdisciplinary recommendation strategy. This allows the model to dynamically adjust recommendation weights based on the interdisciplinary requirements of teaching activities.

(1) Dimension matching score calculation. For a given teaching activity A , its STEAM dimension requirement vector is D_A . For candidate work i , its STEAM dimension label vector is L_i . The dimension matching score between the two is calculated as follows:

$$M_{A,i} = \frac{D_A \cdot L_i}{\|D_A\|_1 + \varepsilon} \quad (14)$$

denotes the dot product, $\|D_A\|_1$ represents the number of demand dimensions, and ε is the smoothing factor to prevent division by zero.

(2) Fusion recommendation score. Combine the dimension matching score with the user preference score to obtain the final recommendation score:

$$S_{u,i} = \alpha \cdot \widehat{y_{u,i}} + (1 - \alpha) \cdot M_{A,i} \quad (15)$$

$\widehat{y_{u,i}}$ represents the predicted user preference score for the output, while α denotes the balancing coefficient (set to 0.7 in this study), which adjusts the weighting between personalized preferences and interdisciplinary teaching objectives.

(3) Group recommendation mechanism. For collaborative learning scenarios where students undertake projects in groups, the model employs a group preference fusion strategy. For a group G consisting of K members, the group preference score is calculated as:

$$\widehat{y_{G,i}} = \frac{1}{K} \sum_{k=1}^K \widehat{y_{u_k,i}} + \beta \cdot Var(\{\widehat{y_{u_k,i}}\}) \quad (16)$$

The first term represents the mean of member preferences, the second term denotes the variance of preferences (greater variance indicates larger differences in interests among members), and the third term is the adjustment coefficient. Additionally, the model supports role-based resource matching, recommending differentiated supplementary materials based on the responsibilities of different roles within the group, such as technical specialists and artistic specialists. The pseudocode for this model is shown in Fig. 3.

Design and analysis of music teaching activities supported by STEAM

When designing music teaching activities based on the idea of STEAM education, this study categorizes project-based learning into four progressive stages and designs differentiated model recommendation strategies for each stage, enabling technology to deeply empower the entire teaching process, as shown in Fig. 4.

(1) Launch Phase: Contextual Introduction and Interest Stimulation.

During the project launch phase, teaching activities aim to spark student interest and cultivate problem awareness. Teachers guide students to explore interdisciplinary themes such as “Music and Technology” and “The Science of Sound,” establishing preliminary research directions. At this stage, the personalized recommendation system prioritizes science-popularization resources (e.g., animations on acoustic principles, analyses of musical instrument sound generation) and contextual introduction materials (e.g., interdisciplinary music project case studies, artist interviews). Resource selection emphasizes science and engineering dimensions to help students develop an interdisciplinary perspective. The recommendation strategy primarily focuses on dimensional matching (with α set to 0.4), de-emphasizing individual preferences to encourage exploration of unfamiliar domains.

(2) Exploration Phase: In-Depth Inquiry and Principle Comprehension.

```
# Transformer Attention Mechanism (Replacement)
def apply_transformer_attention(features):
    # Reshape: Flatten the spatial dimensions into a sequence
    # Input shape: (Batch, Height, Width, Channels)
    b, h, w, c = features.shape
    # Reshape to (Batch, Sequence_Length, Embedding_Dim)
    sequence_input = tf.reshape(features, [b, h * w, c])

    # Multi-Head Self-Attention (MHSA)
    # Allows the model to jointly attend to information from different representation subspaces
    attention_output = layers.MultiHeadAttention(num_heads=4, key_dim=c)(sequence_input, sequence_input)

    # Add & Norm (Residual Connection + Layer Normalization)
    x = layers.Add()([sequence_input, attention_output])
    x = layers.LayerNorm(epsilon=1e-6)(x)

    # Global Average Pooling (Aggregate sequence to vector)
    attended_vector = layers.GlobalAveragePooling1D()(x)

    return attended_vector
```

Fig. 3. CNN-Transformer algorithm applied to the pseudo-code flow chart of personalized music learning recommendation.

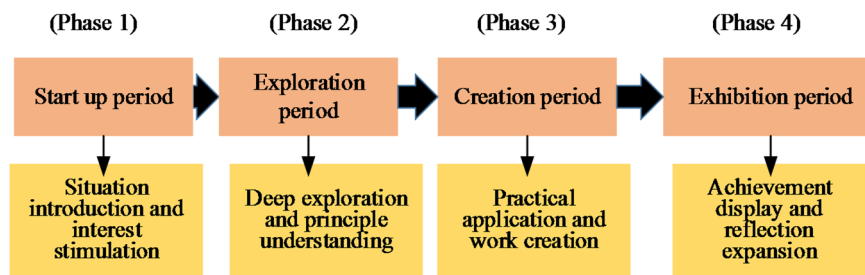


Fig. 4. A four-stage model for music instruction supported by STEAM principles.

During this phase, students conduct deep research around their project theme, learning relevant interdisciplinary knowledge. For example, in the “Creating Music Using Mathematical Principles” project, students must understand mathematical ratios in harmony and fractional relationships in rhythm. The recommendation system now prioritizes technical analysis resources (e.g., audio processing tutorials, introductory music programming guides) and principle-based resources (e.g., mathematical foundations of harmony, physical principles of harmonic series). Resource selection emphasizes technical and mathematical dimensions. The recommendation strategy increases personalized weighting (with α set to 0.6), inferring student interests based on browsing behavior during the initiation phase to achieve “interest-guided deep exploration.”

(3) Creation Phase: Practical Application and Artistic Production.

During the creation phase, students apply acquired knowledge to produce musical works or interdisciplinary projects. For instance, in the “Electronic Music Composition” project, students must integrate audio technology knowledge with musical expression. Here, the recommendation system prioritizes exemplary works (e.g., outstanding student compositions, professional electronic music pieces) and creative tools (e.g., DAW software tutorials, sound design guides), selecting resources based on technical and artistic dimensions. Simultaneously, the system provides personalized creative material recommendations based on students’ technical preferences and artistic inclinations demonstrated during the exploration phase—recommending complex sound synthesis tutorials for technically inclined students and emotional expression analysis case studies for artistically inclined students. The recommendation strategy balances personalization with instructional objectives (with α set to 0.5).

(4) Presentation Phase: Outcome Exhibition and Reflective Expansion.

During the presentation phase, students showcase project outcomes, receive feedback, and engage in reflective expansion. At this stage, the recommendation system emphasizes resources for performance appreciation (e.g., professional concert recordings, interdisciplinary art exhibitions) and evaluation feedback (e.g., work assessment criteria, self-reflection guides), ensuring multidimensional integration in resource selection. The recommendation strategy increases the weight of dimensional matching (with α set to 0.4), encouraging students to engage with a broader range of interdisciplinary art forms and expand their aesthetic horizons.

Beyond project phase segmentation, the model supports group recommendation mechanisms for collaborative learning scenarios. When students undertake projects in groups, the system considers individual preferences and calculates group preferences, recommending comprehensive resources suitable for collaboration, such as ensemble pieces and case studies of interdisciplinary team projects. To illustrate this integrated pathway concretely, this study presents a complete case study of the “Music and Technology” interdisciplinary project. In this project, student teams create a work integrating electronic music with acoustic principles. During the initiation phase, the system recommends contextual resources like the science video “The Physics of Sound” and “The History of Electronic Music” to establish an interdisciplinary perspective. Upon entering the exploration phase, the system provides differentiated recommendations based on team member roles: For the technical lead, it suggested Max/MSP programming tutorials and sound synthesis technology analyses; for the artistic lead, it recommended case studies of electronic music compositions. During the creation phase, the system leveraged prior exploration activities to recommend reference works aligned with the project’s style and specific technical solutions. In the presentation phase, it provided videos showcasing outstanding similar projects and professional evaluation criteria to support student reflection and refinement.

Experimental design and performance evaluation

Datasets collection and experimental environment

To evaluate the performance of the constructed model, this study developed a comprehensive dataset encompassing specific music education scenarios. The raw data was collected from three major online music learning platforms: China University music course section (MOOC), NetEase Cloud Music (education section), and Flowkey (piano learning platform). The collection period spanned from September 2023 to March 2024. The data covers learning resources for four specialized fields: Piano Performance, Vocal Performance, Composition and Composition Technique Theory, and Music Education.

This experiment recruited 112 music majors from a conservatory as participants, including 84 undergraduates and 28 graduate students. Their specializations were distributed as follows: Piano Performance (32), Vocal Performance (28), Music Education (26), and Composition and Composition Technique Theory (26). Participants ranged in age from 18 to 26 years ($M = 21.3$, $SD = 2.1$), comprising 48 males and 64 females. All participants possessed at least three years of professional music training and were familiar with operating online

learning platforms. They were randomly assigned to either the experimental group ($n=58$, using the proposed recommendation system) or the control group ($n=54$, using traditional teaching methods). No significant differences existed between groups in age, gender, or major distribution. All participants voluntarily joined the experiment and signed informed consent forms.

(1) Behavioral log data. Collected user interactions on the learning platform, including page clicks, browsing duration (in seconds), comment content and frequency, as well as resource bookmarking and download records. A total of 12,347 raw interaction records were collected, yielding 5,917 valid records after cleaning and deduplication.

(2) Music resource data. Audio files of teaching pieces and corresponding MIDI/XML score files were collected, totaling 1,031 works. An annotation for each piece was completed by three graduate students in music under faculty supervision. Annotations included: Technical Difficulty Level: Rated on a 1–5 scale (1 = Beginner, 5 = Advanced), referencing the China Musicians Association's Social Art Level Examination textbook standards. Harmonic Complexity: Categorized into simple, moderate, and complex levels. Stylistic Period: Baroque, Classical, Romantic, and Modern periods. Musical Form: Solo pieces, concertos, art songs, opera excerpts, etc. Core Technical Requirements: Such as rapid scales, wide-span arpeggios, polyphonic playing, emotional expression, etc.

(3) STEAM dimension tags. Annotated by the research team using expert review methodology. Each piece was independently annotated by two annotators for relevance across the five STEAM dimensions (0 = Irrelevant, 1 = Relevant). Inter-rater reliability reached 87.3% (Cohen's $\kappa=0.82$), with discrepancies resolved by a third expert (a professor specializing in music education). Dimensions include: Science: Involves acoustic principles, vibration frequency, harmonic series, and other physical-acoustic knowledge; Technology: Covers electronic music, digital audio processing, music programming, and other technical applications; Engineering: Relates to instrument construction, sound field design, recording engineering, and engineering thinking; Art: Encompasses musical expressiveness, aesthetic interpretation, emotional expression (included by default in all works); Mathematical Dimension: Involves mathematical principles such as rhythmic proportions, harmonic mathematical relationships, and formal symmetry.

(4) Psychological scale data. Standardized psychological scales were used to collect psychological state data from experimental and control group students, administered once before the experiment (pre-test) and once after the experiment (post-test). The Kenny Music Performance Anxiety Scale measured stage anxiety levels, the General Self-Efficacy Scale assessed learning confidence, and a self-developed Learning Anxiety Questionnaire evaluated daily practice anxiety. Psychological data were correlated with behavioral log data (click counts, viewing duration, and comment counts) to establish empirical links between behavioral indicators and psychological constructs. This methodology drew upon the research framework proposed by Liu et al. (2023)⁴¹, systematically linking observable learning behaviors with underlying cognitive psychological states. Psychological data collection yielded 112 valid questionnaires: 58 from the experimental group (using the recommendation system) and 54 from the control group (using traditional teaching methods). No significant differences existed between groups in age, grade level, or major background.

After cleaning, deduplication, and standardization, the final dataset comprised 5,917 user interaction records and multimodal feature data from 1,031 music teaching pieces. Specifically: (1) Audio data preprocessing standardized all audio files to a sampling rate of 22.05 kHz, converted them into 128×128-pixel Mel-spectrograms, and applied a Hamming window with a frame length of 1024 samples and a frame shift of 512 samples. (2) Symbolic data preprocessing standardized MIDI files to 480 ticks per quarter note, extracting features such as note sequences, chord progressions, and tempo changes; MusicXML files were parsed into structured musical notation representations. (3) Psychological data preprocessing applied Z-score normalization to questionnaire scores. (4) Randomly split the dataset into training (80%) and testing (20%) sets using seed 42 to ensure reproducibility. The training set contains 4,734 interaction records and 825 compositions, while the testing set includes 1,183 interaction records and 206 compositions. The distribution of behavioral categories remains consistent across both sets.

The hardware configuration of this experimental environment includes a high-performance graphics processing unit (GPU) to speed up model training. TensorFlow is used to realize the RegCycleGAN algorithm in a software framework, and Pandas and NumPy are used for data processing. The specific experimental environment information is shown in Table 1.

Parameters setting

In this experimental parameter configuration, for the CNN feature extraction module: The number of convolutional layers is set to 3, with each layer employing a 3×3 convolution kernel to capture local texture features from the music spectrogram. The rectified linear unit (ReLU) activation function is selected to accelerate

Environmental type	Configuration description	Environmental type	Configuration description
Processor	Intel Core i9-11900 K	Operating system	Windows 10
Internal storage	64GB DDR4 RAM	Programming language	Python 3.8
Memory	2 TB NVMe SSD	Database	MySQL 8.0
GPU	NVIDIA GeForce RTX 3080 (10GB)	Data processing library	Pandas 1.3.3, NumPy 1.21.2
Machine learning framework	TensorFlow 2.2	Input-output interface	Matplotlib

Table 1. Configuration of experimental environment.

convergence and mitigate the vanishing gradient problem. Pooling layers utilize 2×2 max pooling to extract salient features and reduce dimensionality.

For the Transformer sequence modeling module: To capture global dependencies, the number of heads for MHSA is set to 4, and the feature embedding dimension (i.e., the number of CNN output channels) is set to 64. The hidden layer dimension of the feedforward network is set to 128. To prevent overfitting, Dropout is introduced after the self-attention layer and the fully connected layer, with a dropout rate of 0.3.

Training Strategy: The Adam optimizer is selected with an initial learning rate of 0.001. To optimize the training process, a learning rate decay strategy is adopted, halving the learning rate every 20 epochs. The batch size is set to 32, and the total number of iterations (Epochs) is set to 100. The cross-entropy loss function is employed.

Performance evaluation

To comprehensively and objectively evaluate the performance of the proposed CNN-Transformer hybrid model, this study designed rigorous benchmark comparison experiments and ablation studies.

First, the proposed model was compared with the following four mainstream state-of-the-art algorithms to validate its superiority in the music teaching resource recommendation task:

Standard CNN⁴²: A basic convolutional neural network serving as the baseline model.

ResNet-18⁴³: A deep network incorporating residual connections, representing the advanced capabilities of pure CNN architectures.

Pure Transformer⁴⁴: A model relying solely on self-attention mechanisms, used to test pure sequence modeling capabilities.

Chen et al. (2024): The latest relevant research in this field serves as the existing state-of-the-art (SOTA) benchmark.

Firstly, the comparison results of the accuracy, recall, and F1 values of this algorithm with other models are analysed, as shown in Figs. 5, 6 and 7.

In Figs. 5, 6 and 7, by comparing the prediction accuracy results of personalized music learning recommendation, it shows that with the increase of iteration period, the accuracy, recall, and F1 results of each algorithm show a trend of increasing first and then becoming stable. Compared with other model algorithms, the accuracy of the personalized recommendation model based on the CNN-Transformer algorithm model reaches 95.76%, which is at least 4.54% higher. And the prediction accuracy of the personalized music learning recommendation system is the best, followed by the algorithm proposed by Chen et al. (2024), and the prediction accuracy of the Standard CNN algorithm is the lowest. Further analysis of recall and F1 results shows that recall and F1 of this model algorithm reach 90.28% and 92.41%, respectively, which is more than 6% higher than other model algorithms. Therefore, with the support of the personalized music learning recommendation system based on CNN-Transformer, the design of music teaching activities based on the idea of STEAM education can be personalized for user behaviour, with better prediction accuracy.

Additionally, this study introduced an ablation variant (CNN-Base w/o Transformer), which removed the MHSA module from the model while retaining only the CNN backbone, to validate the necessity of the hybrid architecture. Detailed comparison results of the accuracy, recall, and F1-score for each algorithm model on the test set are presented in Table 2.

Further, the teaching activity design of the recommendation system supported by different algorithms and the analysis of user behaviour indicators under traditional teaching methods are shown in Fig. 8.

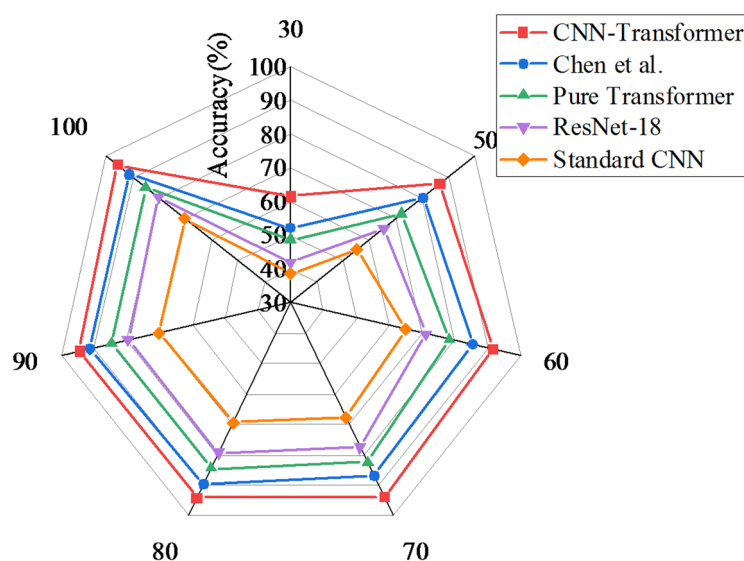


Fig. 5. The Accuracy result chart of personalized music learning recommendation prediction under different algorithms.

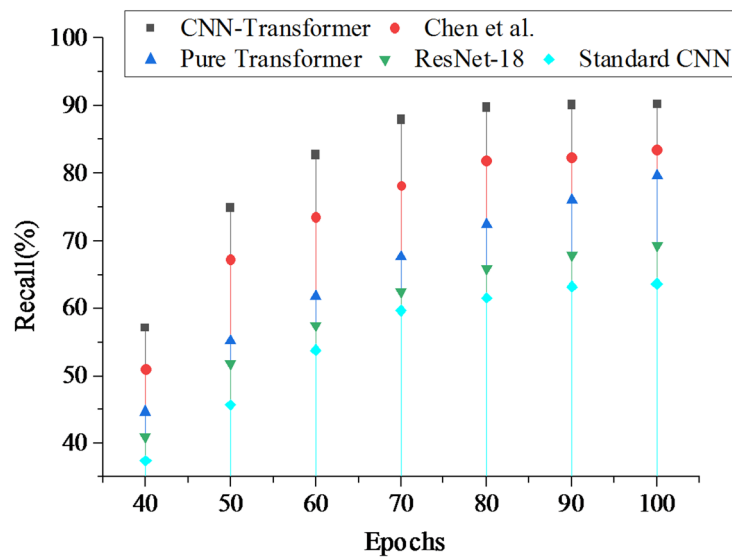


Fig. 6. The Recall result chart of personalized music learning recommendation prediction under different algorithms.

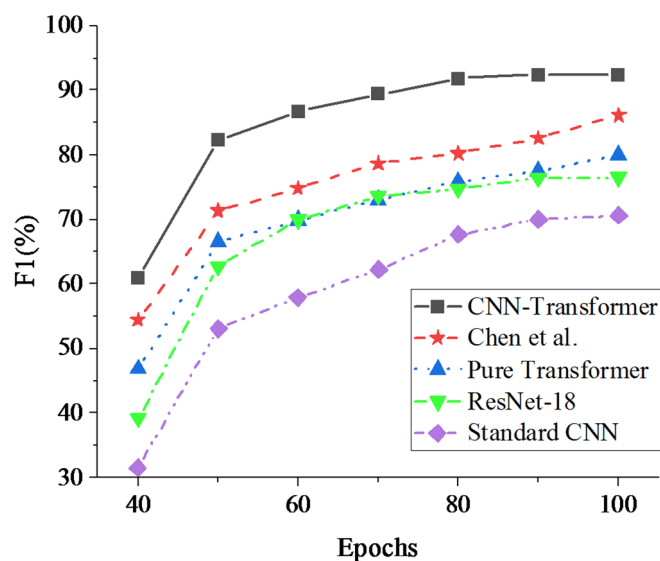


Fig. 7. The F1 result chart of personalized music learning recommendation prediction under different algorithms.

Model algorithm	Accuracy (%)	Recall (%)	F1-Score (%)
Standard CNN	70.22	63.63	70.57
ResNet-18	80.24	69.35	76.52
Pure Transformer	85.11	79.67	79.99
Chen et al. (2024)	91.22	83.57	86.14
CNN-Base (Ablation)	80.15	74.20	77.05
Ours (CNN-Transformer)	95.76	90.28	92.41

Table 2. Benchmark comparison of different models on the test set.

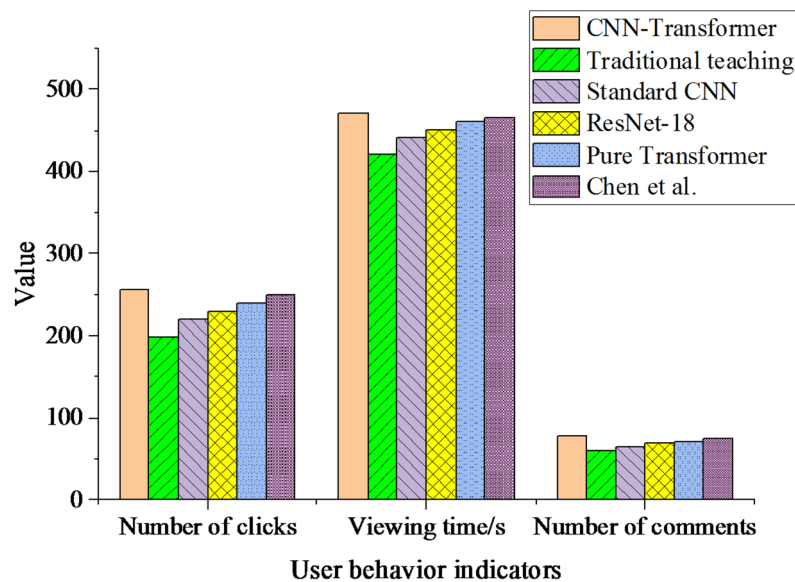


Fig. 8. Results of user behaviour indicators under different teaching methods.

CNN convolutional layer count	Accuracy (%)	Recall (%)	F1-Score (%)	Training time (seconds/epoch)
1 layer	87.32	81.45	83.67	23
2 layers	92.45	86.78	89.12	31
3 layers	95.76	90.28	92.41	42
4 layers	95.21	89.65	91.78	58
5 layers	94.58	88.92	91.03	79

Table 3. The impact of CNN convolution layer depth on model accuracy.

As shown in Fig. 8, the CNN-Transformer model proposed in this study demonstrates significant advantages across all interaction metrics, optimally empowering teaching practices. Specifically, compared to traditional teaching models, the number of page views under this model surged from 198 to 256—a 29.3% increase—while comment interactions rose significantly from 60 to 78. This clearly demonstrates the substantial boost in student engagement driven by precise recommendations. Notably, in rigorous benchmark comparisons against state-of-the-art algorithms, although Chen et al. (2024) and Pure Transformer also demonstrated strong competitiveness (with 250 and 240 clicks, respectively), the CNN-Transformer model maintained leadership across all metrics. Notably, it achieved the highest browsing duration of 471 s, outperforming Chen et al.’s 466 s and the Standard CNN’s 441 s. This comprehensive lead across engagement, retention, and interaction dimensions powerfully demonstrates that hybrid architectures incorporating self-attention mechanisms can more precisely match students’ personalized interests and learning needs. This effectively stimulates their intrinsic motivation and deep learning aspirations in STEAM educational activities, validating the model’s practical value in enhancing teaching effectiveness.

Beyond benchmark comparisons and ablation studies, this study also conducted a systematic parameter sensitivity analysis to examine the impact of core hyperparameters—such as the number of CNN convolutional layers and Transformer attention heads—on model performance. This comprehensive evaluation aims to assess model stability and identify optimal configurations. While keeping the number of Transformer attention heads fixed at 4, this study varied the number of CNN convolutional layers to 1, 2, 3, 4, and 5, observing corresponding changes in model accuracy on the test set. The results are presented in Table 3.

Table 3 results indicate that as the number of convolutional layers increases from 1 to 3, the model accuracy improves from 87.32% to 95.76%, demonstrating that appropriately increasing convolutional layers helps the model extract richer local audio features. When the number of convolutional layers was further increased to 4, the accuracy slightly decreased to 95.21%, and further dropped to 94.58% at 5 layers. This suggests that excessively deep convolutional networks may lead to overfitting or vanishing gradient issues. Therefore, this study selected a 3-layer CNN as the optimal configuration.

Next, while keeping the number of CNN convolutional layers fixed at 3, the number of Transformer attention heads was set to 1, 2, 4, 8, and 16, respectively, to examine the resulting changes in model accuracy on the test set. The results are shown in Table 4.

Table 4 results show that when the number of attention heads increased from 1 to 4, the model accuracy rose from 91.45% to 95.76%, indicating that the multi-head mechanism can capture global dependencies in music

Number of heads	Accuracy (%)	Recall (%)	F1-Score (%)	Training time (seconds/epoch)
1 head	91.45	85.67	87.89	28
2 heads	93.67	88.23	90.45	35
4 heads	95.76	90.28	92.41	42
8 heads	95.68	90.15	92.33	61
16 heads	95.12	89.56	91.78	98

Table 4. The impact of Transformer attention head count on model accuracy.

Number of convolution layers	Number of heads = 1	Number of heads = 2	Number of heads = 4	Number of heads = 8	Number of heads = 16
Number of layers = 1	83.45	85.67	87.32	87.15	86.78
Number of layers = 2	87.23	90.12	92.45	92.21	91.87
Number of layers = 3	91.45	93.67	95.76	95.68	95.12
Number of layers = 4	90.78	93.12	95.21	95.03	94.56
Number of layers = 5	89.34	91.45	94.58	94.12	93.45

Table 5. Joint effect of CNN convolution layers and transformer attention heads (Accuracy: %).

Psychological indicators	Group	Pre-test (M ± SD)	Post-test (M ± SD)	Change amount	Change magnitude	t-value	p-value
Performance anxiety in music	Experimental group (n = 58)	56.3 ± 8.2	48.7 ± 7.6	−7.6	−13.5%	5.67	< 0.001
	Control group (n = 54)	55.8 ± 8.5	54.2 ± 8.3	−1.6	−2.9%	1.23	> 0.05
	Intergroup difference		t = 3.42, p < 0.01				
Self-efficacy	Experimental group (n = 58)	3.2 ± 0.6	3.8 ± 0.5	+ 0.6	+ 18.8%	6.12	< 0.001
	Control group (n = 54)	3.3 ± 0.6	3.4 ± 0.6	+ 0.1	+ 3.0%	1.08	> 0.05
	Intergroup difference		t = 2.98, p < 0.01				
Learning anxiety	Experimental group (n = 58)	45.2 ± 7.1	39.3 ± 6.5	−5.9	−13.1%	4.89	< 0.001
	Control group (n = 54)	44.8 ± 7.3	43.1 ± 7.0	−1.7	−3.8%	1.31	> 0.05
	Intergroup difference		t = 2.67, p < 0.05				

Table 6. Pre-test and Post-test comparison of psychological indicators between the experimental and control groups. Note: Higher scores on the Music Performance Anxiety Scale indicate more severe anxiety. Generally, higher scores on the Self-Efficacy Scale indicate stronger self-efficacy. The self-designed Academic Anxiety Scale uses a 5-point scoring system with a total range of 15–75 points; higher scores indicate more severe academic anxiety. Paired-sample t-tests were used for within-group pre- and post-test comparisons, while independent-sample t-tests were used for between-group post-test comparisons.

sequences from different representational subspaces. When the number of heads increased to 8, the accuracy remained stable (95.68%), while a slight decrease occurred at 16 heads (95.12%). This suggests that excessive attention heads may introduce redundant information, increasing computational complexity with limited performance gains. Therefore, this study selects 4 heads as the optimal configuration.

To further investigate the interaction between the two parameters, this study conducted a grid search experiment, with results shown in Table 5.

As shown in Table 5, the model performance exhibits a “first increase then decrease” trend for both parameters. The optimal combination is 3 convolutional layers and 4 attention heads, achieving an accuracy of 95.76%. This result indicates that moderate network depth and multi-head counts can achieve the optimal balance between local feature extraction and global dependency modeling. Conversely, overly deep or wide network structures may impair model performance. This study also conducted sensitivity analyses on the dropout rate (0.1–0.5) and learning rate (0.0001–0.01). Experiments revealed optimal performance at a Dropout rate of 0.3—higher rates cause underfitting, while lower rates risk overfitting. A learning rate of 0.001 achieves the optimal balance between convergence speed and final performance.

To validate the system’s effectiveness in improving students’ psychological states, this study compared pre- and post-intervention changes in psychological scale scores between the experimental group (using this recommendation system) and the control group (traditional teaching), as shown in Table 6.

The experimental results showed that the music performance anxiety scores of students in the experimental group decreased from 56.3 points (SD = 8.2) before the intervention to 48.7 points (SD = 7.6) after the intervention, representing a reduction of 13.5%. While the anxiety scores of the control group decreased only slightly from 55.8 points (SD = 8.5) to 54.2 points (SD = 8.3), representing a decrease of just 2.9%. An independent samples t-test revealed a significant difference in post-test scores between the two groups (t = 3.42, p < 0.01), indicating

that this system can effectively alleviate students' stage fright. Regarding self-efficacy, the experimental group's score increased from 3.2 (SD = 0.6) pre-intervention to 3.8 (SD = 0.5) post-intervention, representing an 18.8% improvement. The control group's score rose slightly from 3.3 (SD = 0.6) to 3.4 (SD = 0.6), a 3.0% increase. The difference between groups was significant ($t = 2.98, p < 0.01$). Regarding learning anxiety, the experimental group decreased from 45.2 points (SD = 7.1) before the intervention to 39.3 points (SD = 6.5), a reduction of 13.1%; the control group decreased from 44.8 points (SD = 7.3) to 43.1 points (SD = 7.0), a reduction of 3.8%. The difference between groups was significant ($t = 2.67, p < 0.05$).

Correlation analysis between behavioral and psychological indicators revealed that the increase in browsing duration was significantly negatively correlated with the decrease in anxiety scores ($r = -0.42, p < 0.01$). While the increase in click-through rates showed a significant positive correlation with the increase in self-efficacy ($r = 0.38, p < 0.05$). Furthermore, the increase in comment counts exhibited a significant negative correlation with the decrease in learning anxiety ($r = -0.35, p < 0.05$). These findings indicate an intrinsic connection between heightened behavioral engagement on the platform and improved psychological well-being among students, providing preliminary validation for the validity of the "technology-instruction-psychology" research logic.

Discussion

The proposed CNN-Transformer hybrid recommendation system demonstrates superior performance across multiple benchmarks. As shown in Table 2, this model significantly outperforms traditional Standard CNN and ResNet-18 in both accuracy and F1 score. This result corroborates Mu & Wu (2023)'s⁴⁵ perspective that the effective fusion of deep learning architectures is crucial for enhancing recommendation systems' ability to represent complex data. While traditional convolutional networks excel at extracting local spatiotemporal textures from audio, their limited receptive fields often hinder the capture of longer-span phrase structures within musical compositions. By introducing a Transformer module based on Multi-Header Self-Attention (MHSA), this study successfully overcomes this limitation, achieving a unified representation of local features and global context.

Furthermore, Gholamiangonabadi & Grolinger (2023)⁴⁶ emphasized the pivotal role of deep neural networks in constructing highly personalized models. The ablation results from this study further validate this assertion: removing the Transformer module causes the model's accuracy to drop by over 5%. This demonstrates that the self-attention mechanism, by dynamically computing feature weights, not only enhances sequence modeling capabilities but also enables the precise capture of each student's unique musical preferences. Consequently, it surpasses existing generic models in personalized recommendation accuracy.

From the perspective of instructional design and psychological practice, technological breakthroughs in algorithmic precision directly translate into enhanced educational experiences. Experimental data (Fig. 8) show that students using this system significantly outperformed traditional teaching groups and other algorithmic groups in browsing duration (471 s) and interactive comments (78). This marked increase in user engagement aligns closely with findings from da Silva et al. (2023)⁴⁷ in their review of educational recommendation systems: the core value of intelligent recommendation systems lies in stimulating learners' intrinsic motivation and sustained engagement through precise resource matching. Within the STEAM education framework, this high-precision recommendation effectively reduces students' cognitive load when retrieving interdisciplinary resources, enabling them to enter the "flow" state of deep learning more rapidly. By constructing an intelligent learning environment that offers instant responsiveness and contextually relevant content, this system enhances knowledge acquisition efficiency and psychologically strengthens students' self-efficacy and exploratory drive, providing robust technological support for cultivating the comprehensive competencies of music majors.

Conclusion

Research contribution

This study innovatively constructs a CNN-Transformer hybrid recommendation model based on STEAM education principles, effectively addressing the technical limitations of traditional algorithms in capturing long-range dependencies in music. Through rigorous benchmark comparisons, the model demonstrates significantly superior performance over ResNet-18 and state-of-the-art models in accuracy (95.76%), recall (90.28%), and F1 score (92.41%), validating its algorithmic superiority. In instructional design applications, practical data from music teaching activities based on this model show that compared to traditional teaching methods: - Student page clicks increased from 198 to 256, a 29.3% rise; - Browsing duration extended from 421 s to 471 s, an 11.9% increase; - Comment interactions grew from 60 to 78, a 30.0% surge. These significant improvements in behavioral metrics demonstrate that the personalized recommendation system proposed in this study effectively enhances students' behavioral engagement with music learning resources. By providing personalized resource recommendations aligned with students' interests and learning needs, the system encourages more proactive and sustained interaction with the learning platform.

Within the STEAM education framework, this heightened behavioral engagement creates favorable conditions for interdisciplinary exploration. Students devote more time to accessing and experiencing diverse music-related content, laying a foundation for subsequent deep learning. By demonstrating that the recommendation system significantly boosts student behavioral engagement, this study provides a reliable empirical basis for applying educational technology in music instruction. It also establishes a methodological reference for future research exploring the connections between technology and psychological/cognitive dimensions.

Future works and research limitations

It should be noted that this study has significant limitations in terms of dataset scale. The dataset comprises only 1,031 musical works, which is relatively limited for Transformer architectures that typically require large-scale data to effectively model global dependencies. Despite employing Dropout regularization, this sample size still

carries risks of overfitting and may not sufficiently cover the stylistic diversity and technical genres required for professional music education. Therefore, the current model should be regarded as proof-of-concept, primarily demonstrating the feasibility of the CNN-Transformer hybrid architecture for personalized music learning recommendations rather than serving as a mature tool directly applicable to university course instruction.

To address this limitation, future research will adopt two complementary strategies: First, introducing transfer learning based on pre-trained models. Models pre-trained on large-scale music corpora—such as MAESTRO for piano performance, the Lakh MIDI Dataset for notated music, and JSB Chorales for harmonic structures—will initialize the feature extraction module of this study, followed by fine-tuning on specialized instructional data. This approach has been proven effective in mitigating overfitting when training data is limited. Second, expanding the dataset scale by collaborating with multiple music conservatories and educational institutions to collect teaching repertoire data covering broader stylistic periods (Baroque, Classical, Romantic, Modern), technical difficulty levels (beginner to advanced), and cultural traditions. This will enhance the model's generalization capability and practical application value. Although this study preliminarily establishes the association between behavioral indicators and psychological indicators, the mechanisms underlying psychological state improvement require further exploration. Future research will introduce more sophisticated psychometric tools (e.g., physiological monitoring, eye-tracking, facial expression analysis) combined with qualitative methods to deeply uncover how personalized recommendations influence students' cognitive processes. Concurrently, it will explore dynamic recommendation strategy optimization based on psychological states to build a truly “psychologically friendly” intelligent teaching system.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author, Xiaohua Wang, on reasonable request via e-mail 20211036@qhnu.edu.cn.

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Author contributions

Tingting Zhuang: Conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft preparation Xiaohua Wang: writing—review and editing, visualization, supervision, project administration, funding acquisition Dongyun Chang: methodology, software, validation, formal analysis, investigation.

Declarations

Competing interests

The authors declare no competing interests.

Ethics statement

The studies involving human participants were reviewed and approved by the College of Music, Qinghai Normal University Ethics Committee (Approval Number: 2023.012032564). The participants provided their written informed consent to participate in this study. All methods were performed in accordance with relevant guidelines and regulations.

Additional information

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